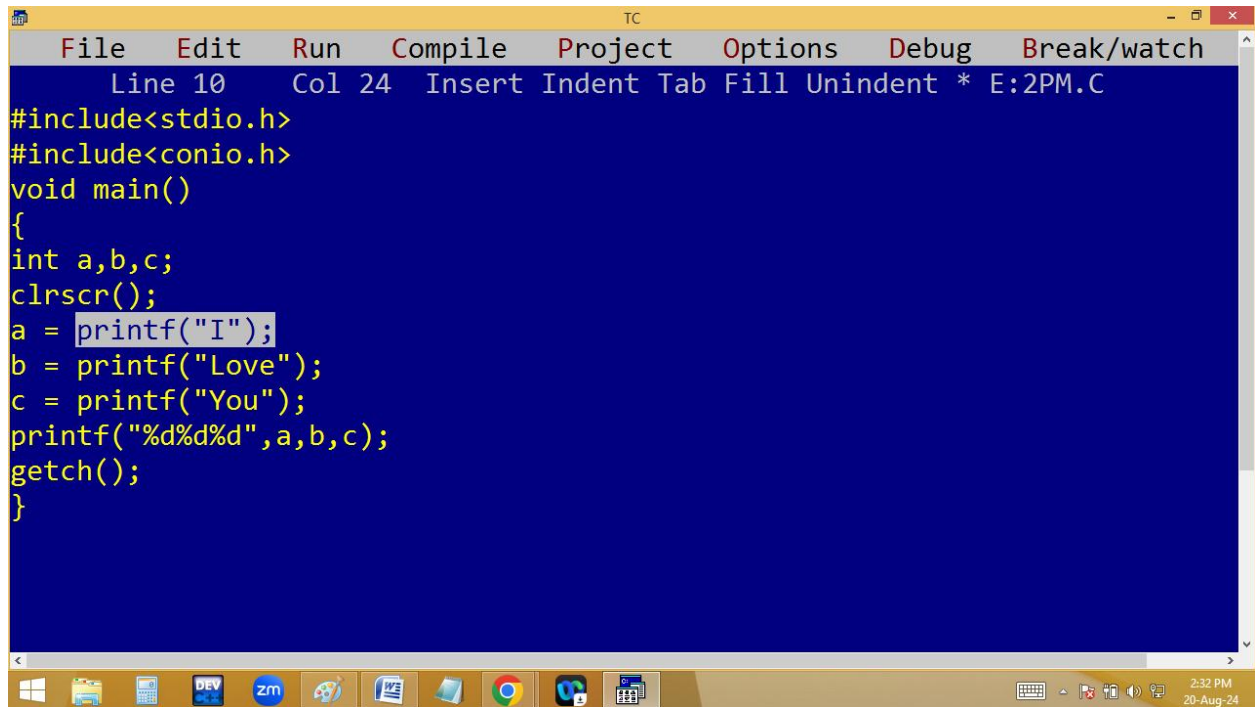


```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 8 Col 9 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n", 5 && -5 );
printf("%d\n", 5.5 && 0 );
printf("%d\n", '0' || 0 );
printf("%d\n", !0 && !-1 );
getch();
}
```

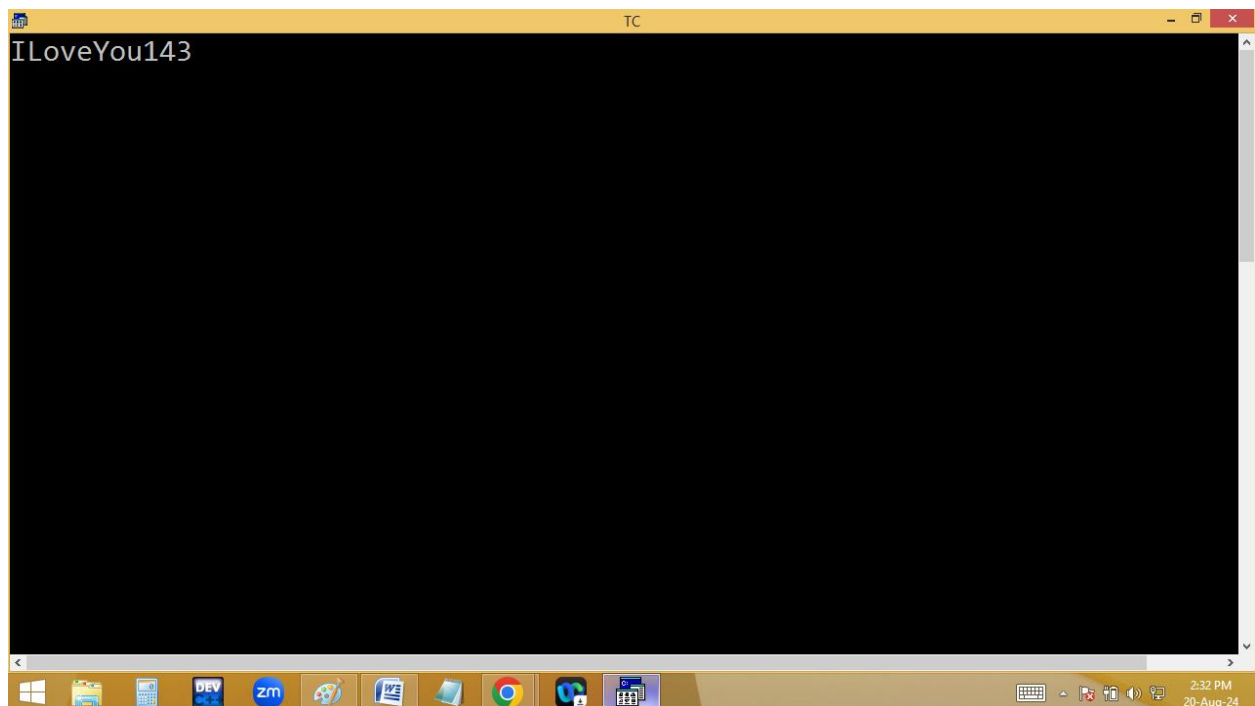
```
TC
1
0
1
0
_
```



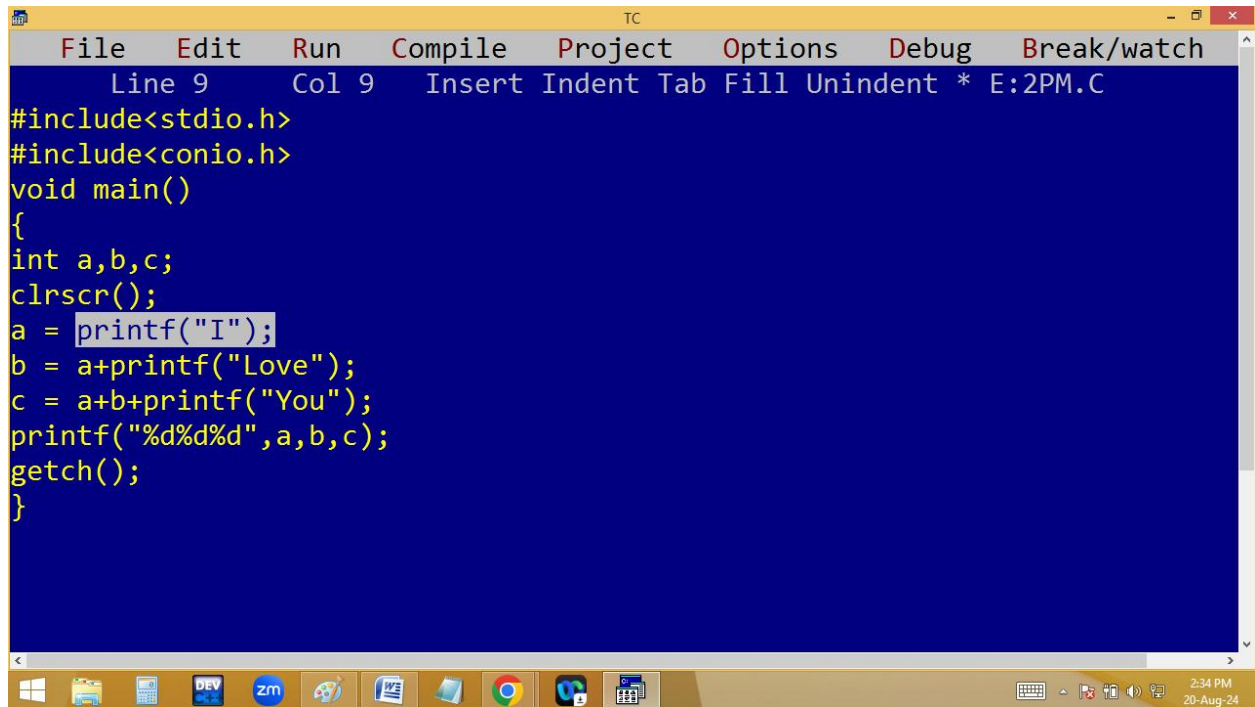
The screenshot shows the Turbo C++ (TC) IDE with a yellow title bar and a menu bar containing File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the top indicates 'Line 10 Col 24 Insert Indent Tab Fill Unindent * E:2PM.C'. The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a,b,c;
clrscr();
a = printf("I");
b = printf("Love");
c = printf("You");
printf("%d%d%d",a,b,c);
getch();
}
```

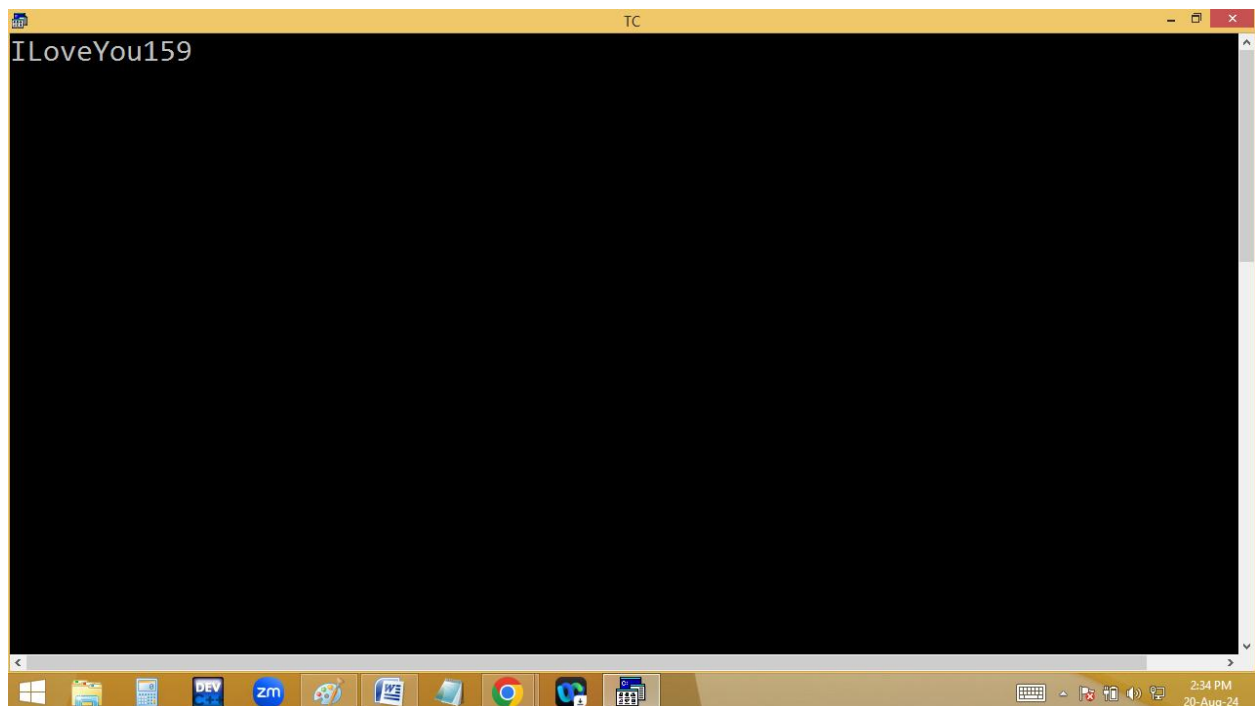
The taskbar at the bottom shows various application icons, including Windows Explorer, DEV, zm, and others, along with system icons for keyboard, network, and volume. The system clock shows 2:32 PM on 20-Aug-24.



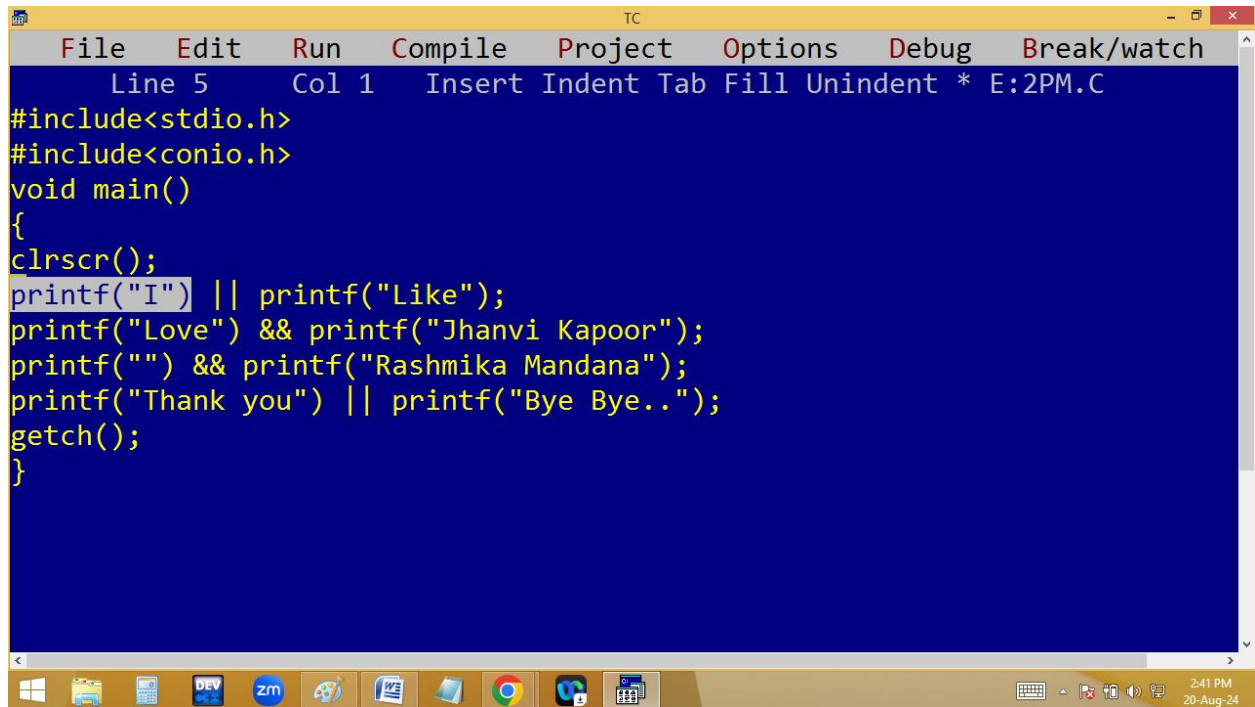
The screenshot shows the Turbo C++ (TC) IDE with a yellow title bar. The main editing area has a black background and displays the output of the program: 'I Love You143'. The taskbar at the bottom is identical to the first screenshot, showing the same application icons and system clock (2:32 PM on 20-Aug-24).



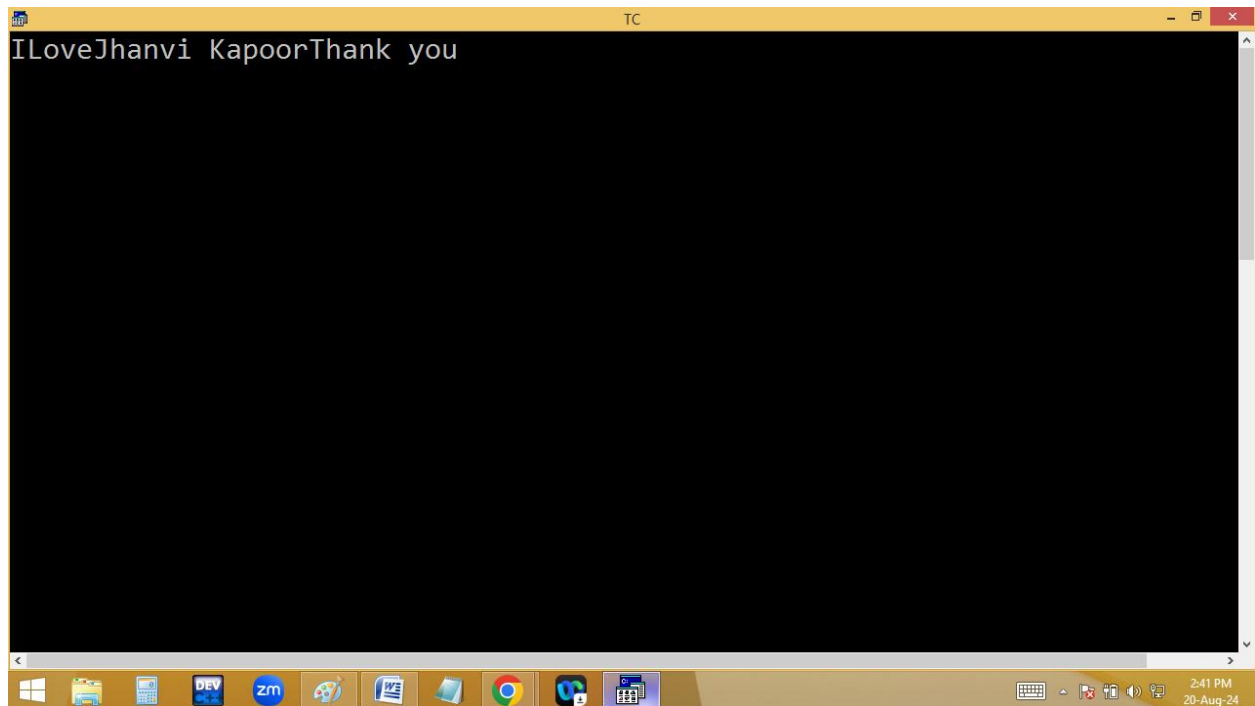
```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 9 Col 9 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a,b,c;
clrscr();
a = printf("I");
b = a+printf("Love");
c = a+b+printf("You");
printf("%d%d%d",a,b,c);
getch();
}
```



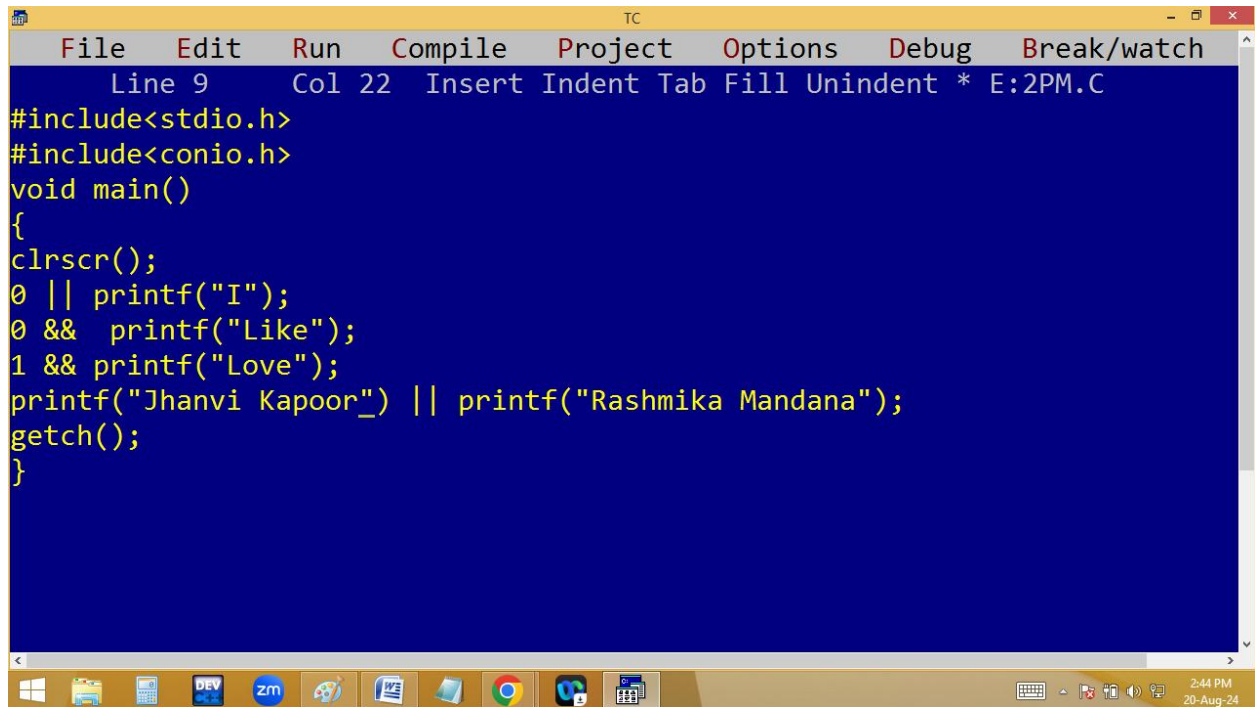
```
TC
I Love You159
2:34 PM
20-Aug-24
```



```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 5 Col 1 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("I") || printf("Like");
printf("Love") && printf("Jhanvi Kapoor");
printf("") && printf("Rashmika Mandana");
printf("Thank you") || printf("Bye Bye..");
getch();
}
```



```
TC
ILoveJhanvi KapoorThank you
```

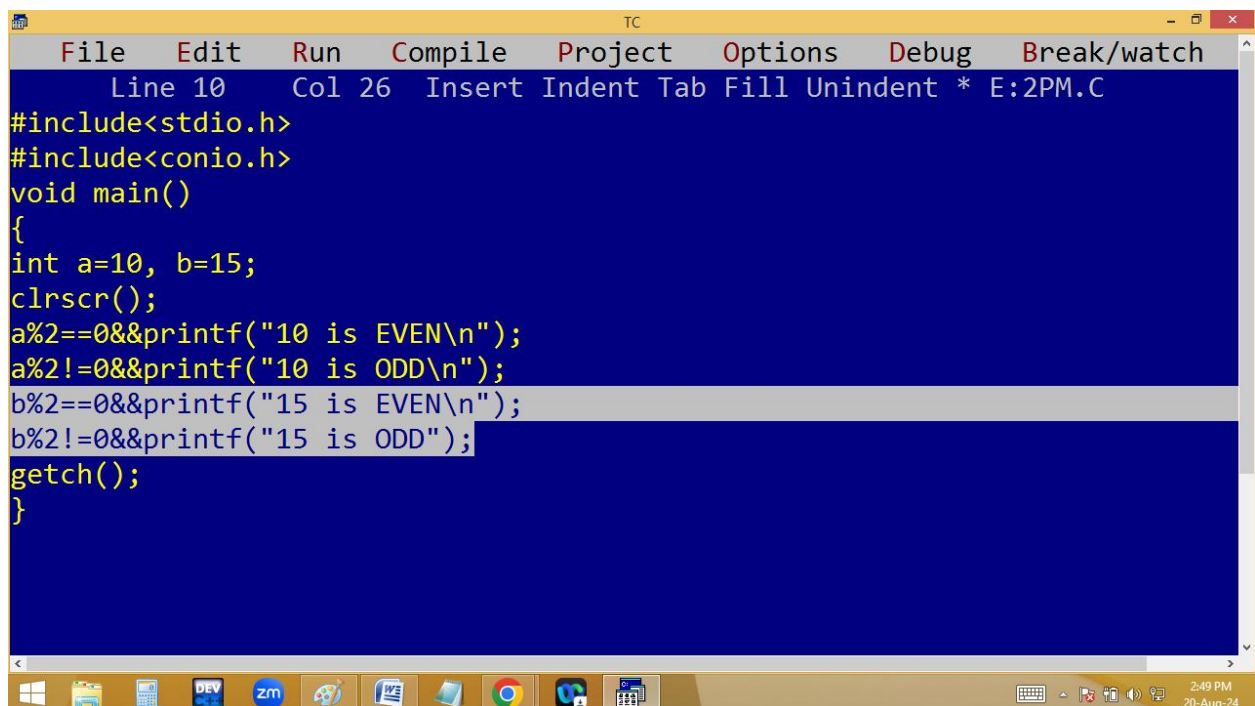


The screenshot shows the Turbo C++ (TC) IDE with a menu bar (File, Edit, Run, Compile, Project, Options, Debug, Break/watch) and a status bar (Line 9, Col 22, Insert, Indent, Tab, Fill, Unindent, * E:2PM.C). The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    clrscr();
    0 || printf("I");
    0 && printf("Like");
    1 && printf("Love");
    printf("Jhanvi Kapoor_") || printf("Rashmika Mandana");
    getch();
}
```

The Windows taskbar at the bottom shows various icons including DEV, zm, and system icons, with the time 2:44 PM and date 20-Aug-24.

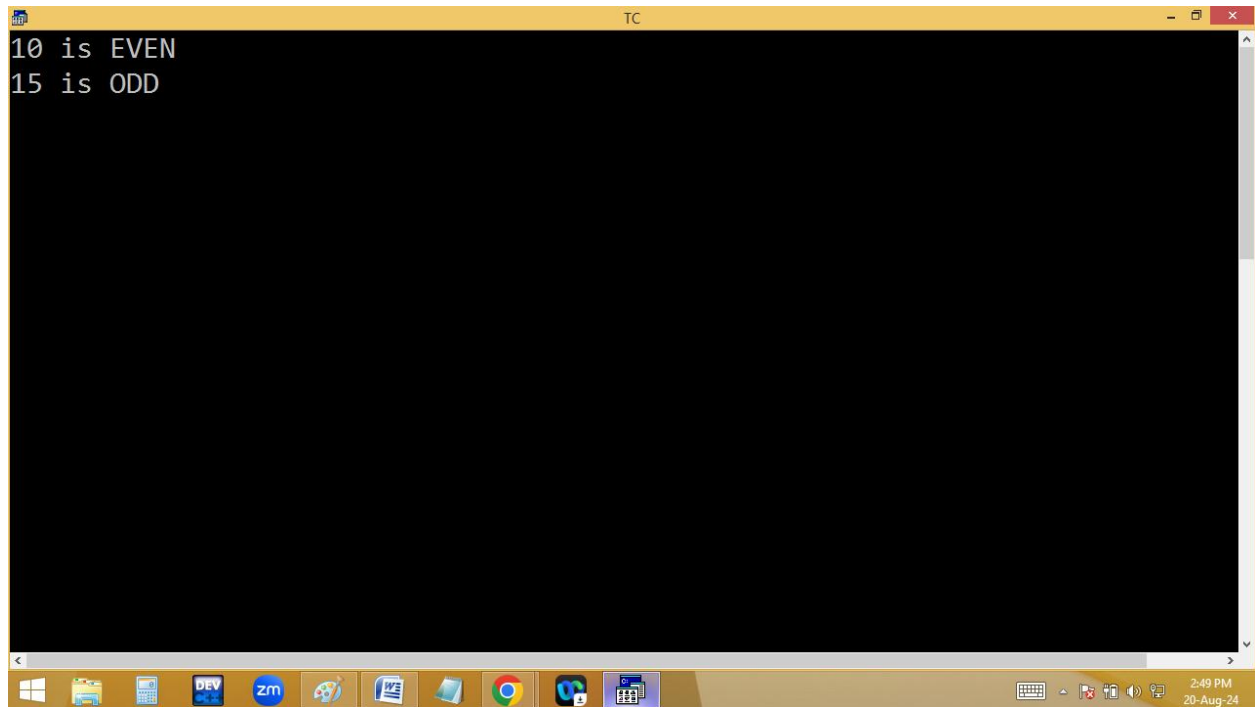
Finding even/odd without using if / ternary operator:



The screenshot shows the Turbo C++ (TC) IDE with a menu bar (File, Edit, Run, Compile, Project, Options, Debug, Break/watch) and a status bar (Line 10, Col 26, Insert, Indent, Tab, Fill, Unindent, * E:2PM.C). The code in the editor is as follows:

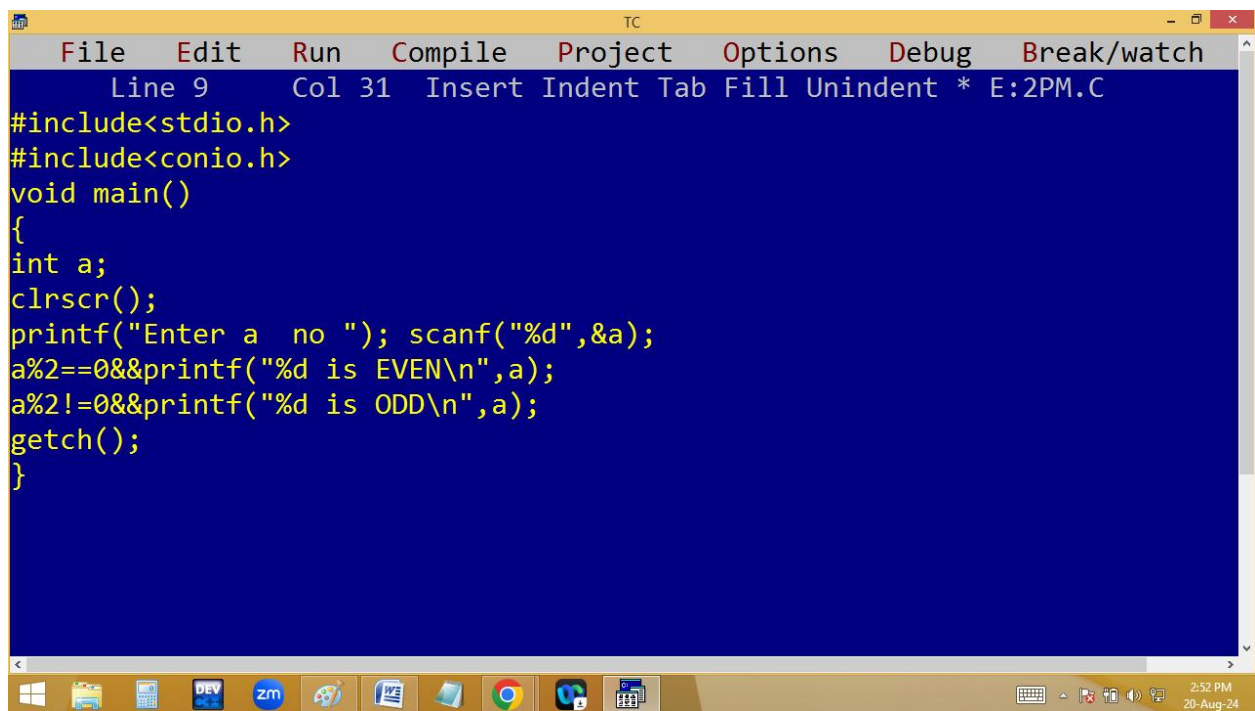
```
#include<stdio.h>
#include<conio.h>
void main()
{
    int a=10, b=15;
    clrscr();
    a%2==0&&printf("10 is EVEN\n");
    a%2!=0&&printf("10 is ODD\n");
    b%2==0&&printf("15 is EVEN\n");
    b%2!=0&&printf("15 is ODD");
    getch();
}
```

The Windows taskbar at the bottom shows various icons including DEV, zm, and system icons, with the time 2:49 PM and date 20-Aug-24.



A screenshot of a Turbo C++ (TC) console window. The window has a yellow title bar with the text "TC" and standard window controls. The main area is black with white text. The text displayed is "10 is EVEN" on the first line and "15 is ODD" on the second line. At the bottom, there is a Windows taskbar with various icons including Windows, File Explorer, DEV, zm, and others. The system clock in the bottom right corner shows "2:49 PM" and "20-Aug-24".

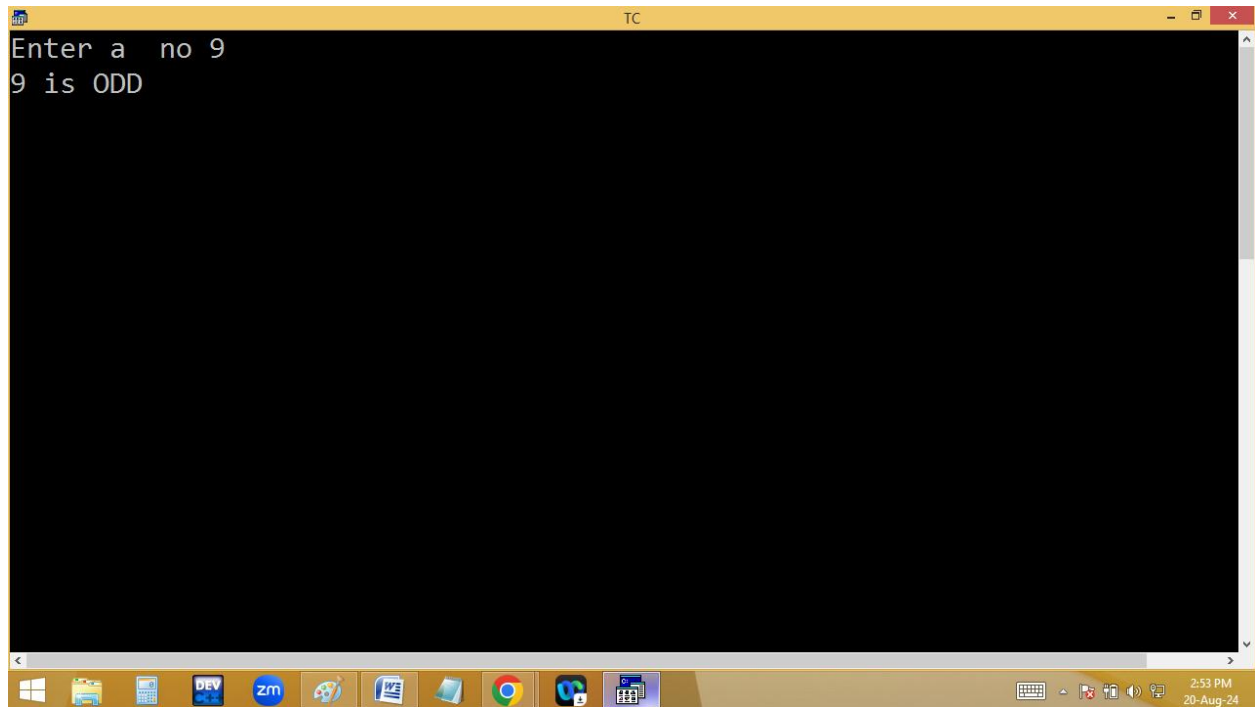
```
10 is EVEN
15 is ODD
```



A screenshot of a Turbo C++ (TC) editor window. The window has a yellow title bar with the text "TC" and standard window controls. Below the title bar is a menu bar with the following items: File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. Below the menu bar is a status bar that reads "Line 9 Col 31 Insert Indent Tab Fill Unindent * E:2PM.C". The main area has a blue background with yellow text. The code displayed is as follows:

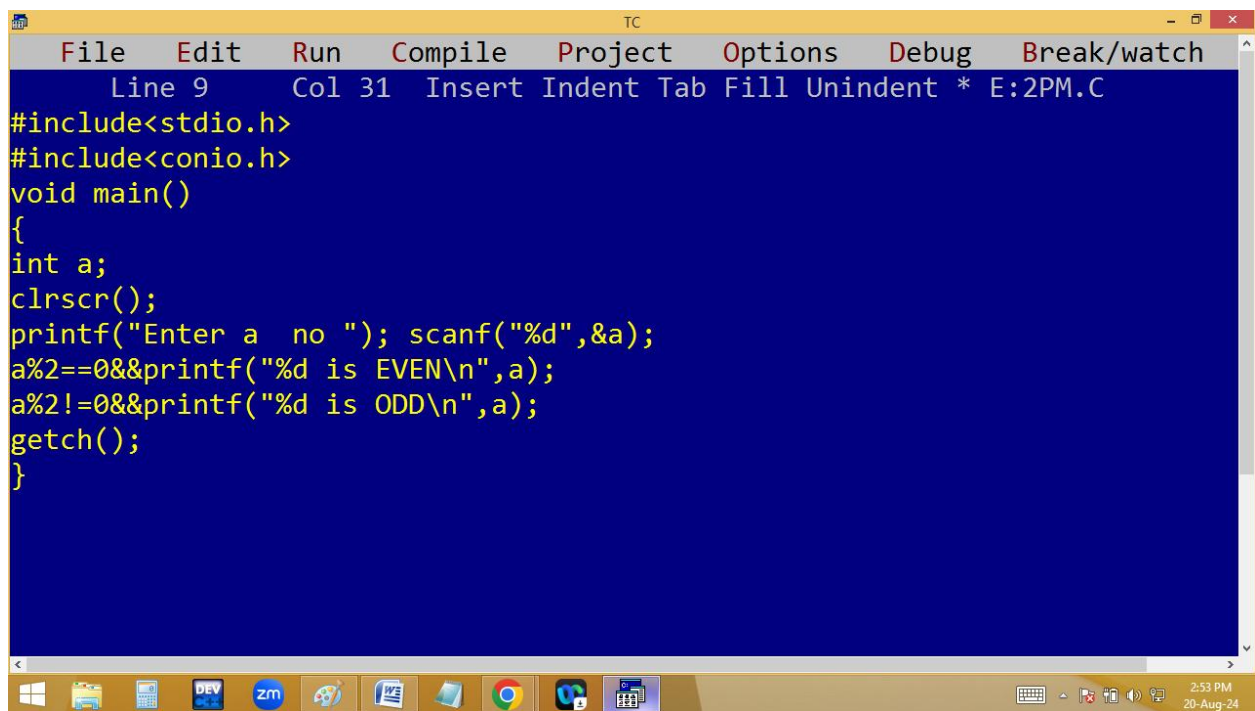
```
#include<stdio.h>
#include<conio.h>
void main()
{
    int a;
    clrscr();
    printf("Enter a no "); scanf("%d",&a);
    a%2==0&&printf("%d is EVEN\n",a);
    a%2!=0&&printf("%d is ODD\n",a);
    getch();
}
```

At the bottom, there is a Windows taskbar with various icons including Windows, File Explorer, DEV, zm, and others. The system clock in the bottom right corner shows "2:52 PM" and "20-Aug-24".



A screenshot of a Turbo C++ (TC) console window. The window has a yellow title bar with the text "TC" and standard window controls. The main area is black with white text. The text displayed is "Enter a no 9" on the first line and "9 is ODD" on the second line. The Windows taskbar is visible at the bottom, showing various application icons and the system clock indicating 2:53 PM on 20-Aug-24.

```
Enter a no 9
9 is ODD
```



A screenshot of a Turbo C++ (TC) editor window. The window has a yellow title bar with the text "TC" and standard window controls. Below the title bar is a menu bar with the following items: File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. Below the menu bar is a status bar that reads "Line 9 Col 31 Insert Indent Tab Fill Unindent * E:2PM.C". The main area has a blue background with yellow text. The code displayed is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a;
clrscr();
printf("Enter a no "); scanf("%d",&a);
a%2==0&&printf("%d is EVEN\n",a);
a%2!=0&&printf("%d is ODD\n",a);
getch();
}
```

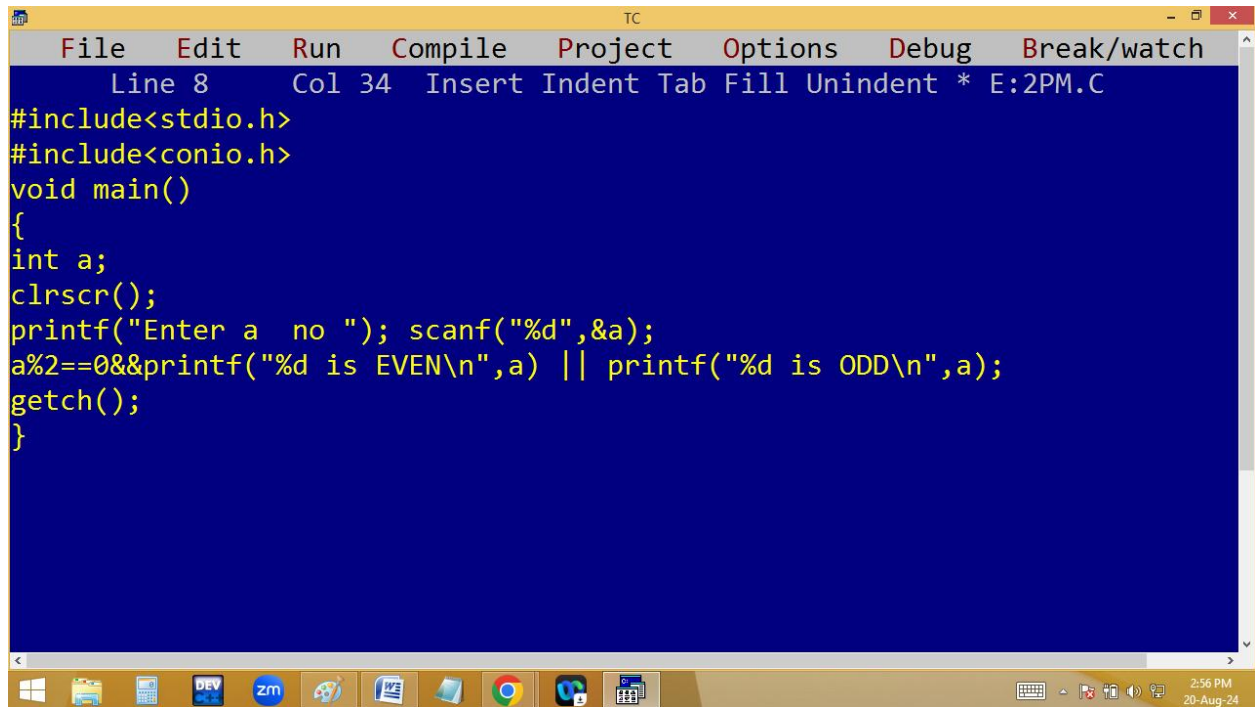
The Windows taskbar is visible at the bottom, showing various application icons and the system clock indicating 2:53 PM on 20-Aug-24.

```
TC
Enter a no 55
55 is ODD
```

A screenshot of a Windows 10 desktop. The taskbar at the bottom shows icons for Windows, File Explorer, Calculator, DEV C++, Zoom, Paint, Word, Edge, Chrome, and a folder icon. The system tray on the right shows the date and time as 2:53 PM on 20-Aug-24. A Turbo C++ (TC) window is open, displaying the text 'Enter a no 55' and '55 is ODD' on a black background.

```
TC
Enter a no 30
30 is EVEN
```

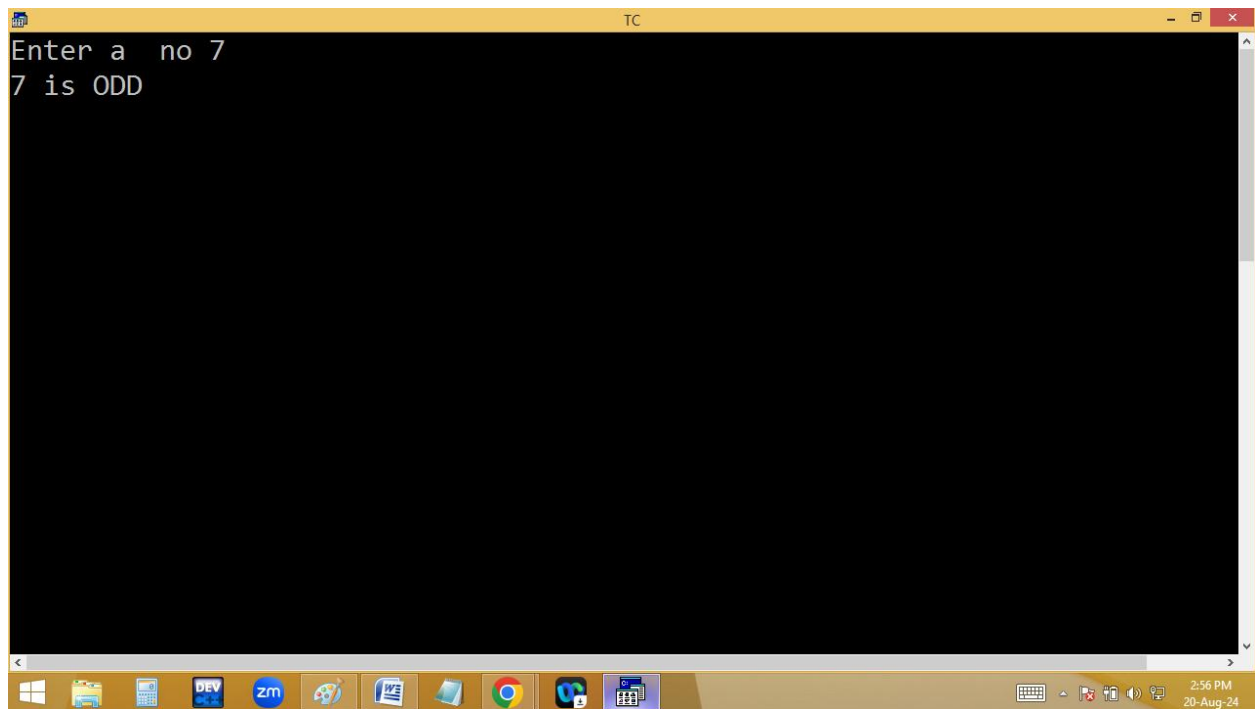
A screenshot of a Windows 10 desktop, similar to the one above. The taskbar and system tray are identical. The Turbo C++ (TC) window is open, displaying the text 'Enter a no 30' and '30 is EVEN' on a black background.



The screenshot shows the Turbo C++ (TC) IDE with a blue background. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the top indicates 'Line 8 Col 34 Insert Indent Tab Fill Unindent * E:2PM.C'. The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a;
clrscr();
printf("Enter a no "); scanf("%d",&a);
a%2==0&&printf("%d is EVEN\n",a) || printf("%d is ODD\n",a);
getch();
}
```

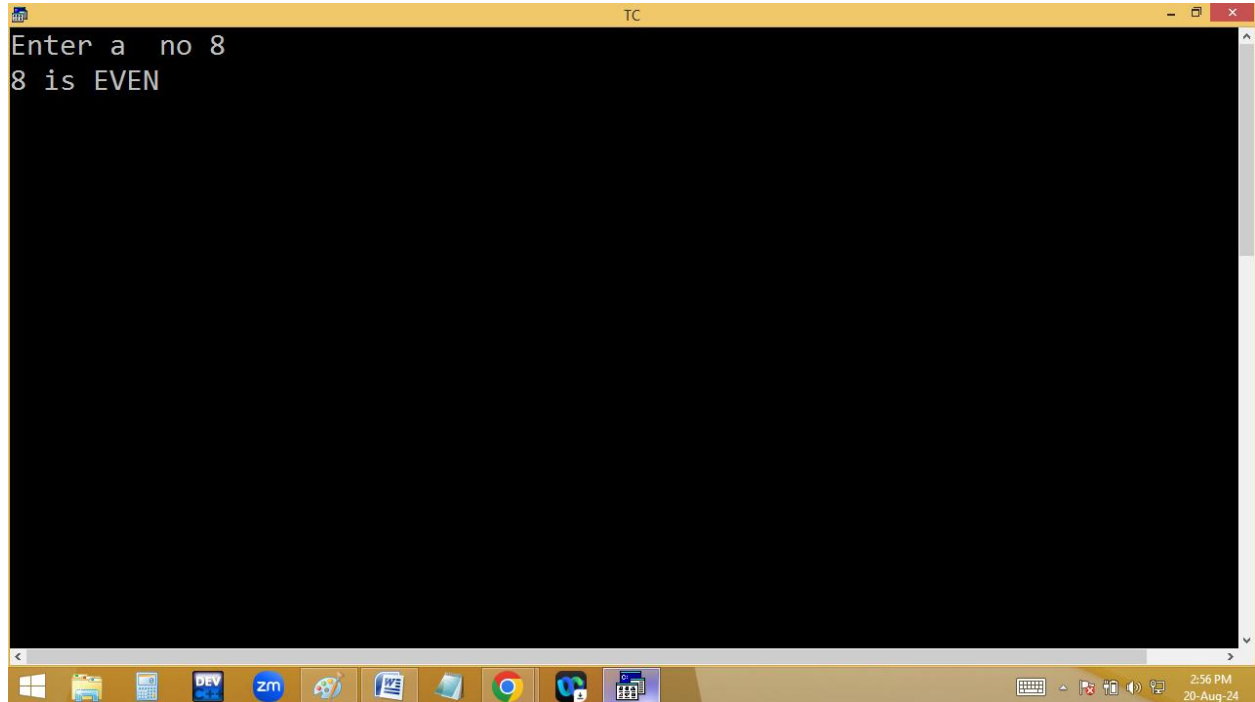
The Windows taskbar at the bottom shows various icons including the Start button, File Explorer, Calculator, DEV C++, Zoho Mail, and others. The system clock in the bottom right corner displays '2:56 PM' and '20-Aug-24'.



The screenshot shows the same Turbo C++ IDE window after execution. The output window at the top displays the following text:

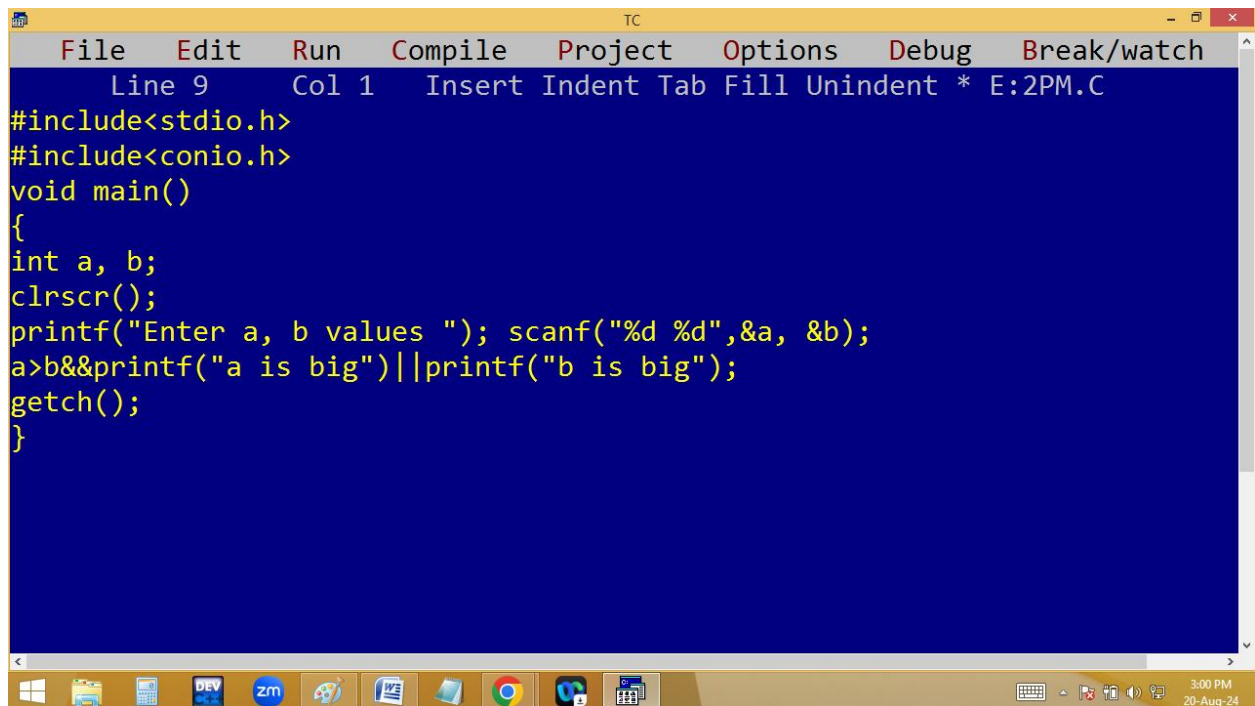
```
Enter a no 7
7 is ODD
```

The rest of the IDE window is black. The Windows taskbar and system clock remain the same as in the first screenshot.



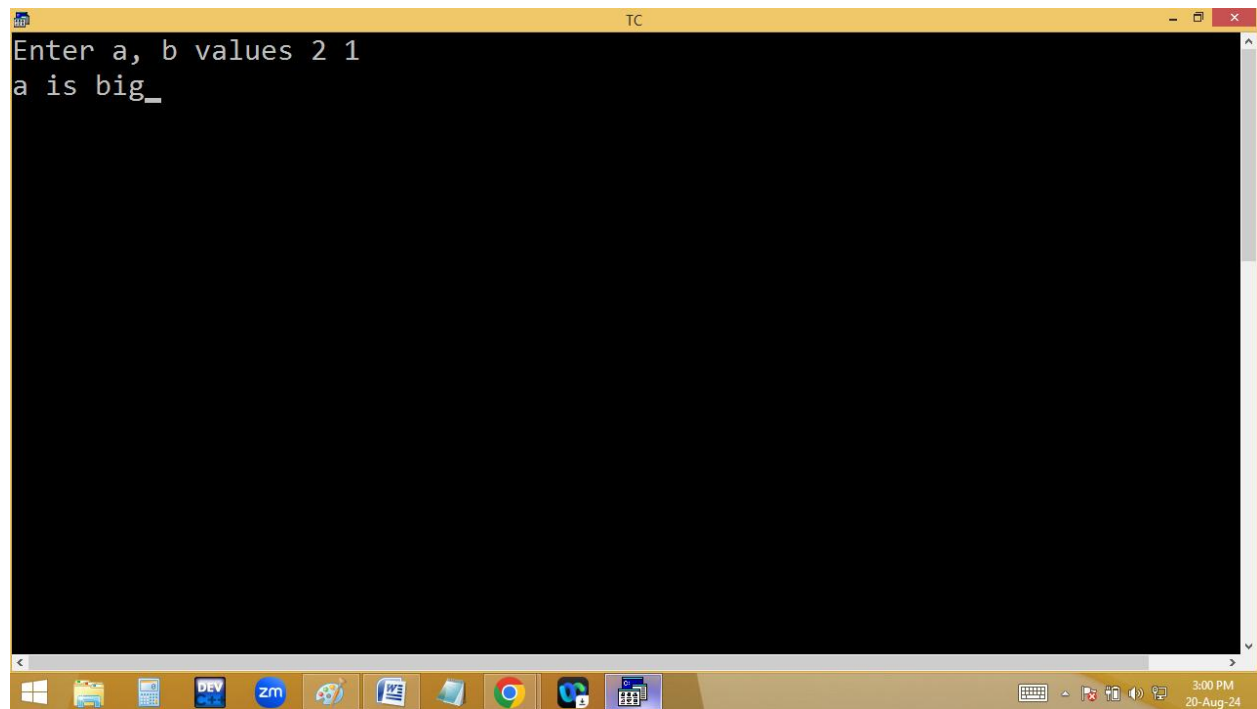
```
Enter a no 8
8 is EVEN
```

Finding max in 2 no's without using if..else / ternary op:

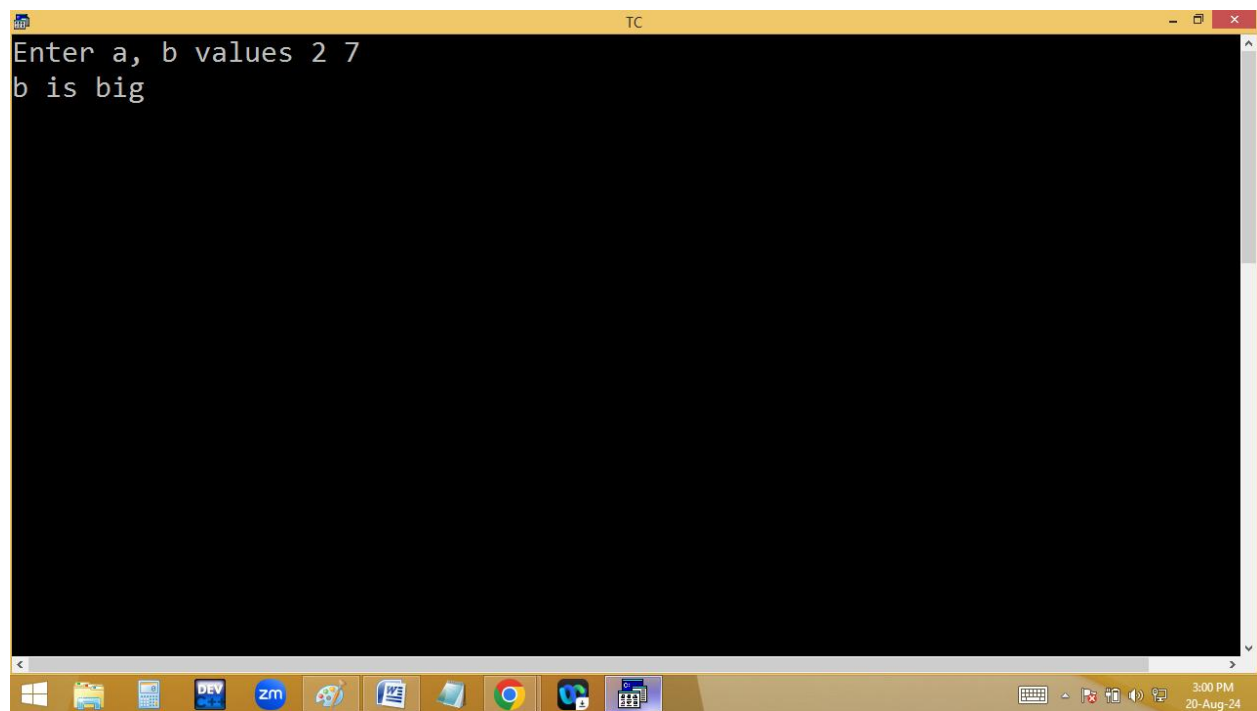


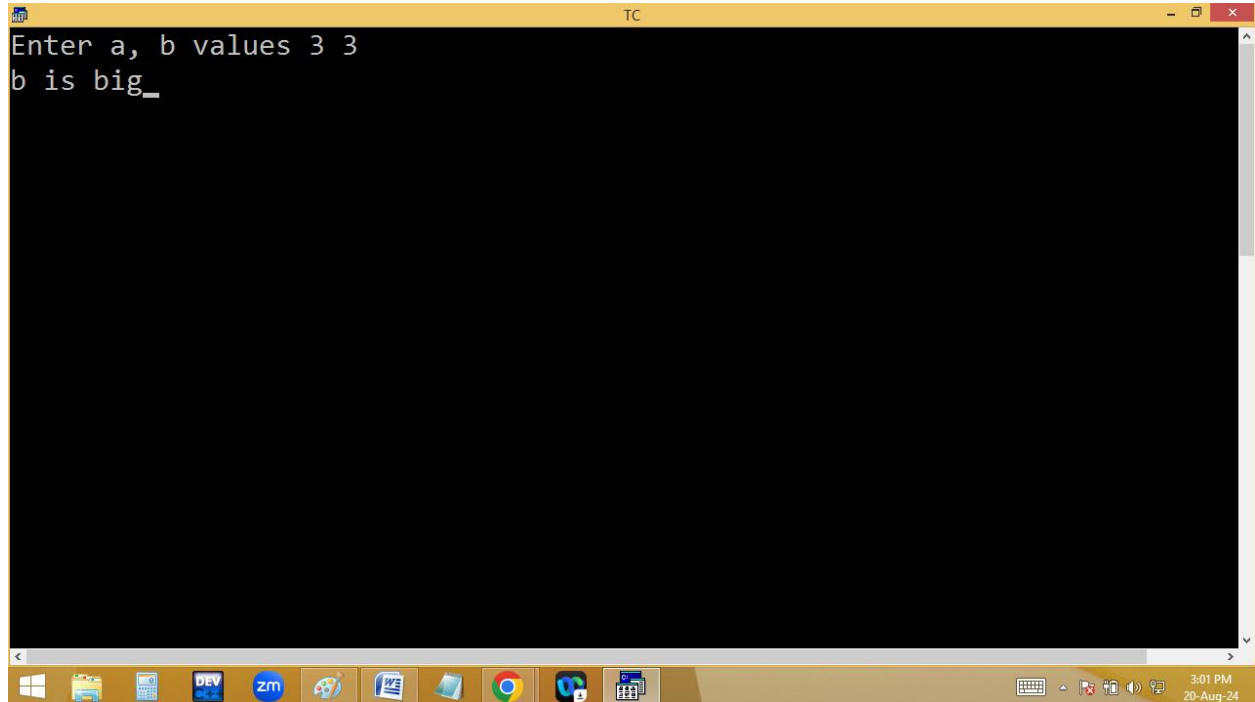
```
File Edit Run Compile Project Options Debug Break/watch
Line 9 Col 1 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a, b;
clrscr();
printf("Enter a, b values "); scanf("%d %d",&a, &b);
a>b&&printf("a is big")||printf("b is big");
getch();
}
```

```
TC
Enter a, b values 2 1
a is big_
```



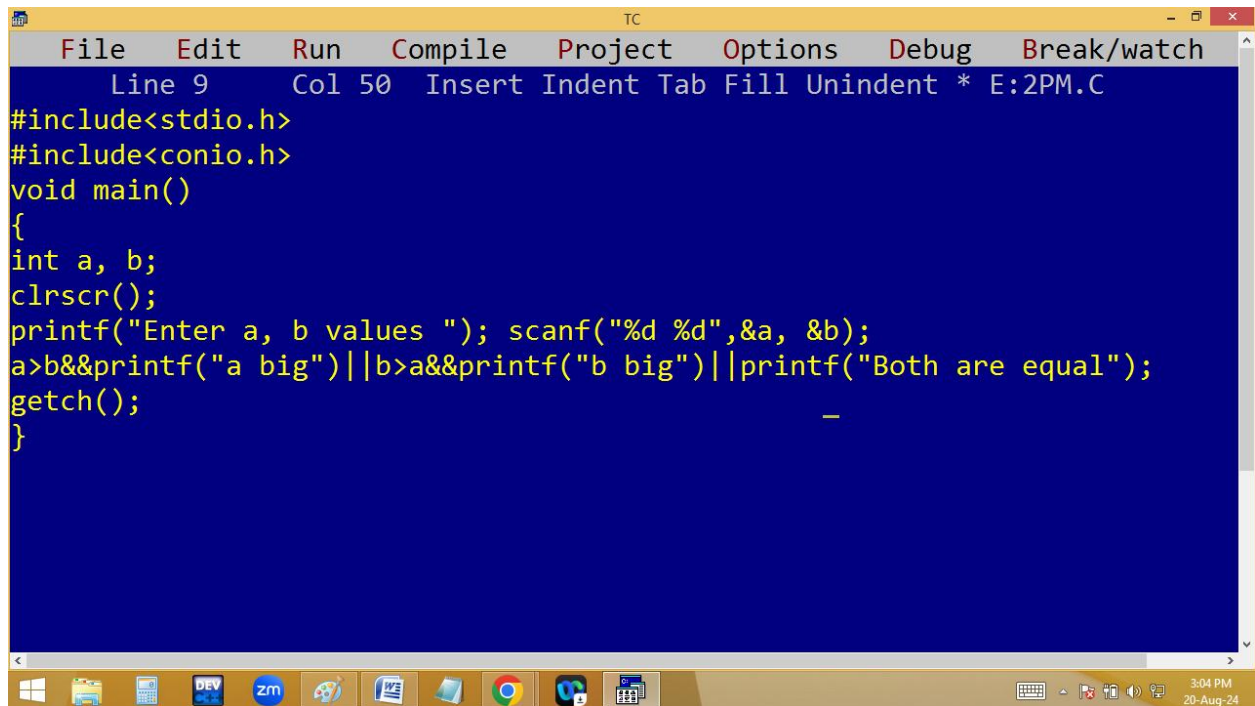
```
TC
Enter a, b values 2 7
b is big
```





A screenshot of a Turbo C++ (TC) console window. The window has a yellow title bar with the text "TC". The main area is black with white text. The text displayed is "Enter a, b values 3 3" on the first line and "b is big_" on the second line. The Windows taskbar is visible at the bottom, showing various application icons and the system clock indicating 3:01 PM on 20-Aug-24.

```
Enter a, b values 3 3
b is big_
```

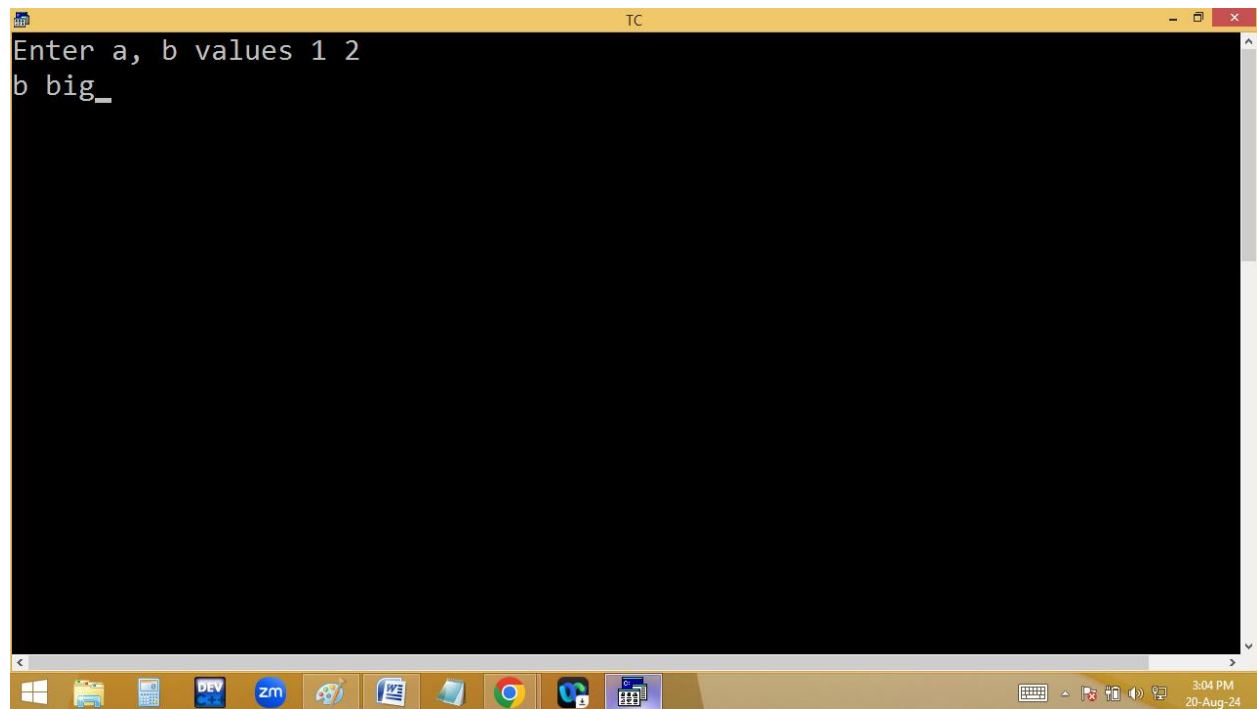


A screenshot of a Turbo C++ (TC) editor window. The window has a yellow title bar with the text "TC". Below the title bar is a menu bar with the following options: File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. Below the menu bar is a status bar that reads "Line 9 Col 50 Insert Indent Tab Fill Unindent * E:2PM.C". The main area is blue with yellow text. The code displayed is as follows:

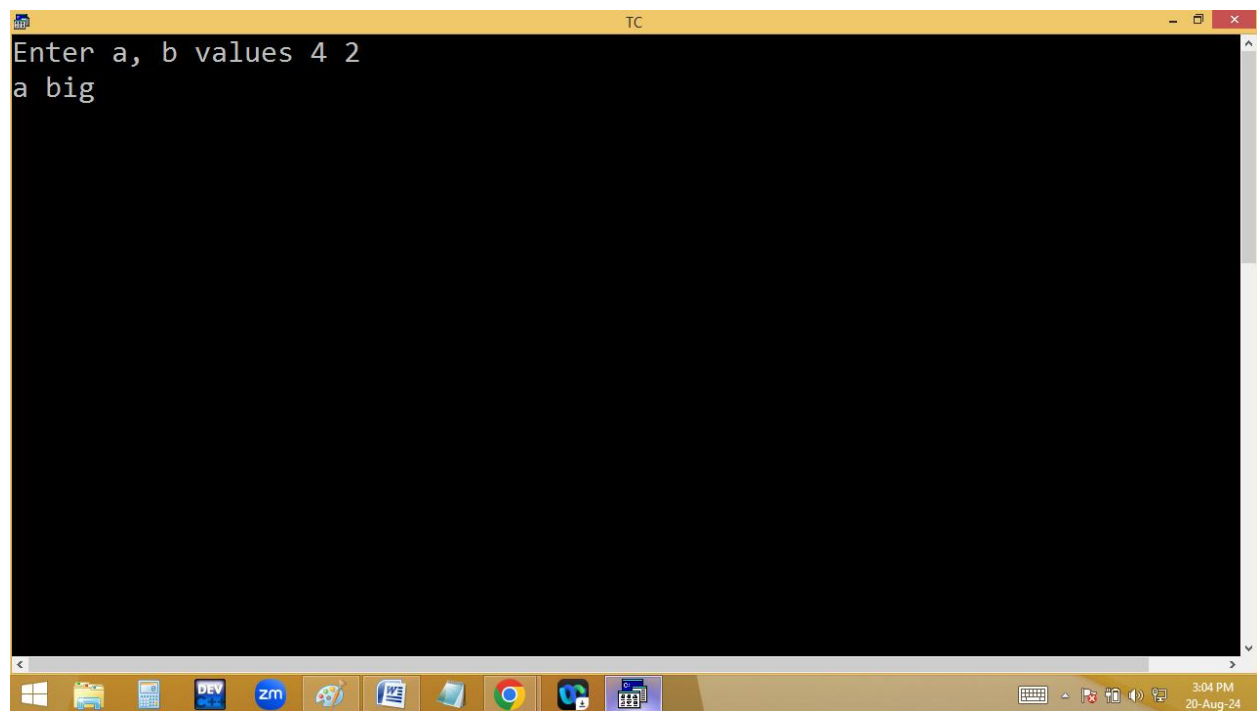
```
#include<stdio.h>
#include<conio.h>
void main()
{
int a, b;
clrscr();
printf("Enter a, b values "); scanf("%d %d",&a, &b);
a>b&&printf("a big")||b>a&&printf("b big")||printf("Both are equal");
getch();
}
```

The Windows taskbar is visible at the bottom, showing various application icons and the system clock indicating 3:04 PM on 20-Aug-24.

```
TC
Enter a, b values 1 2
b big_
```



```
TC
Enter a, b values 4 2
a big
```





```
Enter a, b values 9 9
Both are equal_
```

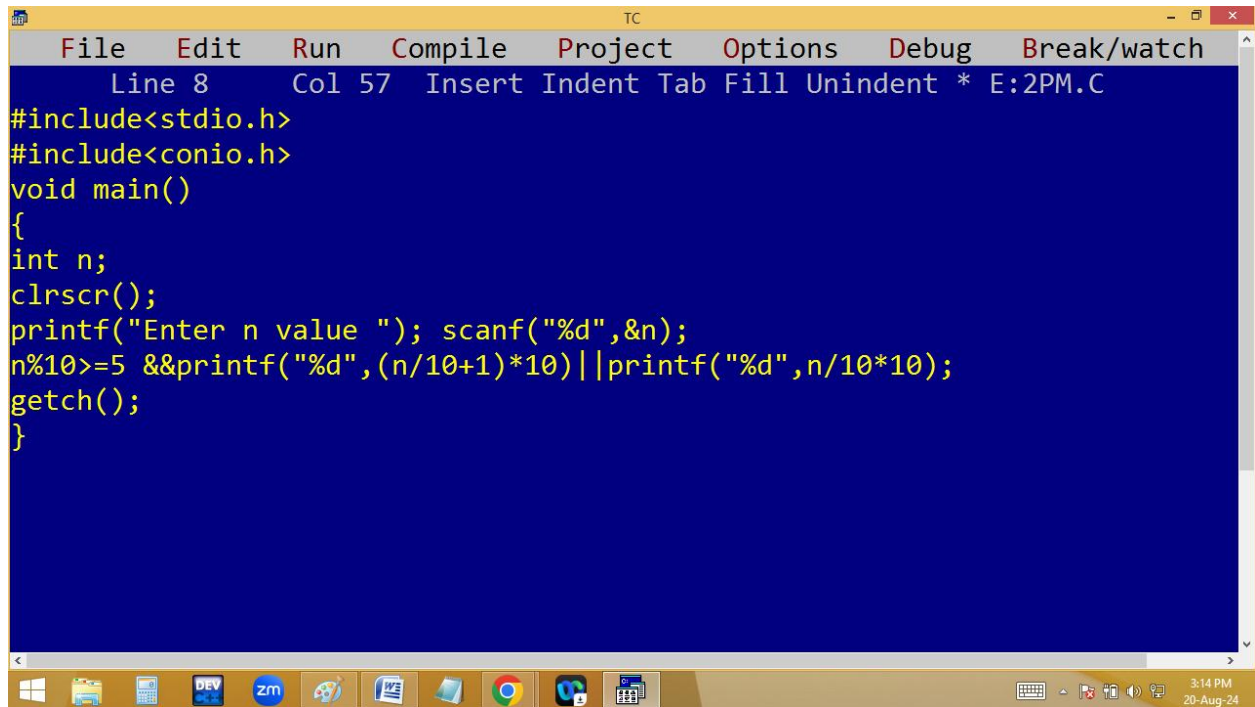
**Rounding of given no to next or previous multiple
without using if..else / ternary operator.**

If the last digit ≥ 5 next multiple of ten

If the last digit < 5 before multiple fo ten

27 $\rightarrow$ last digit is 7 which is $\geq 5 \rightarrow 30$

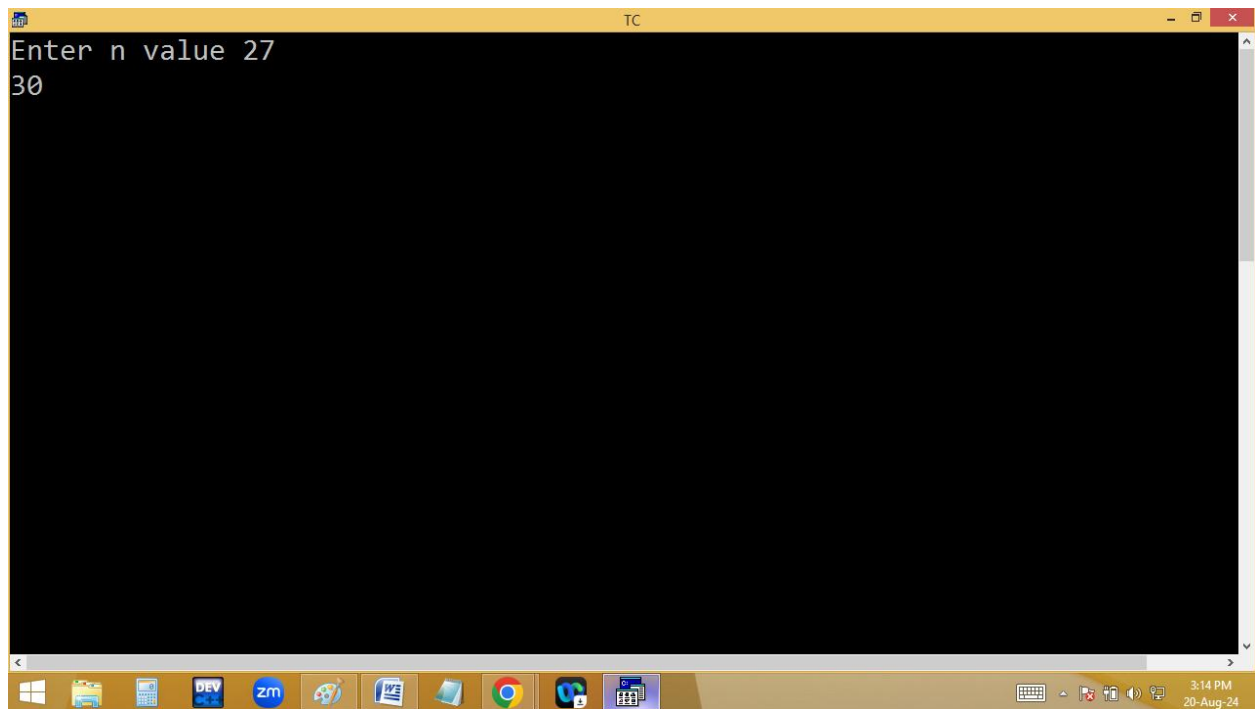
24 $\Rightarrow$ last digit is 4 which $< 5 \rightarrow 20$



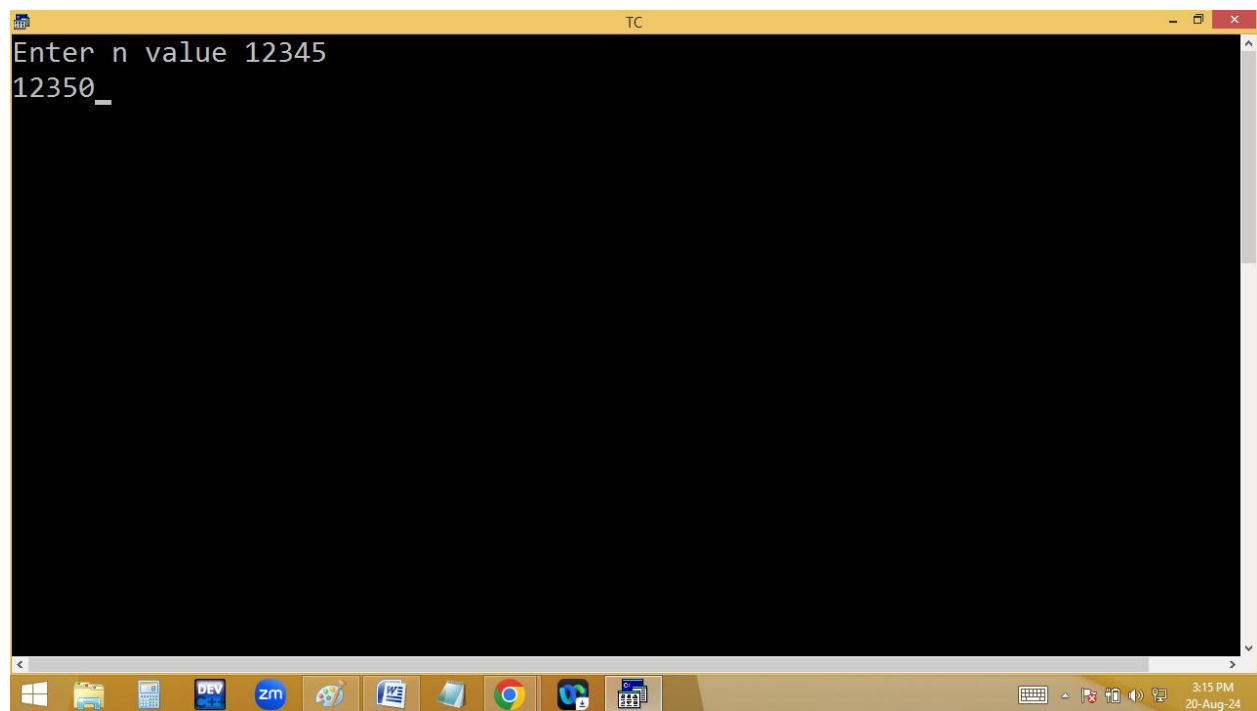
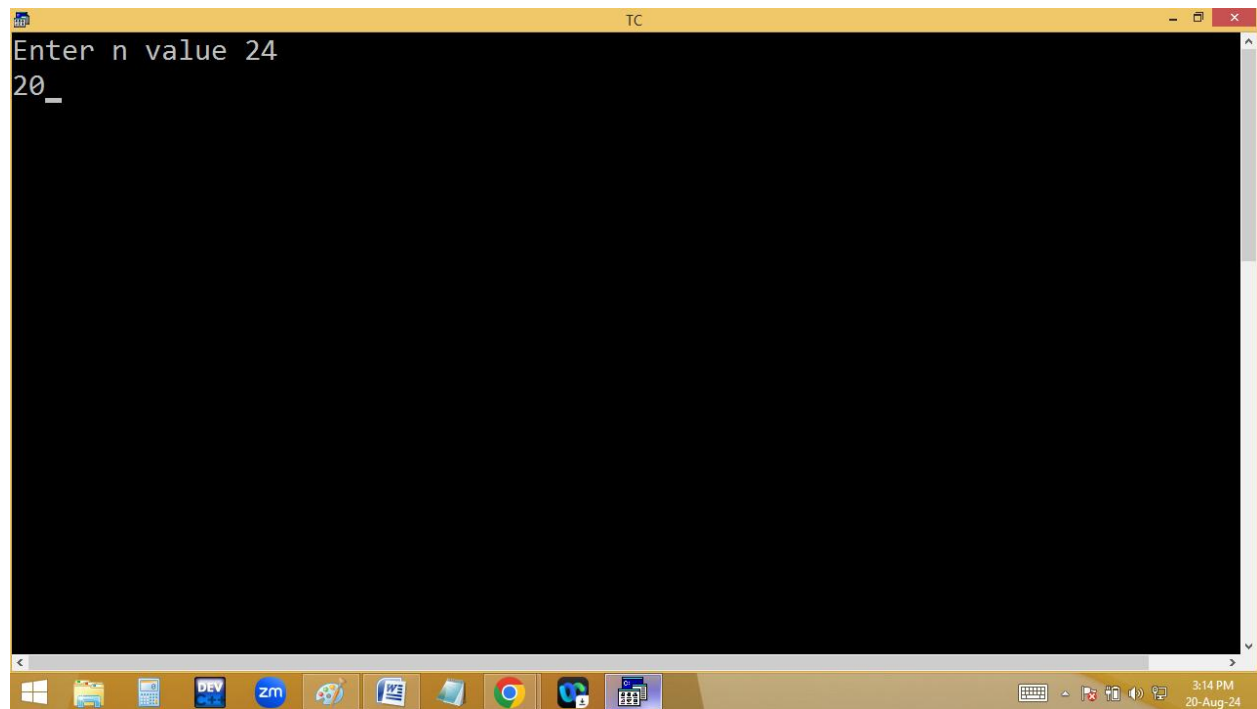
The screenshot shows the Turbo C++ (TC) IDE with a menu bar (File, Edit, Run, Compile, Project, Options, Debug, Break/watch) and a status bar (Line 8, Col 57, Insert Indent Tab Fill Unindent * E:2PM.C). The code in the editor is as follows:

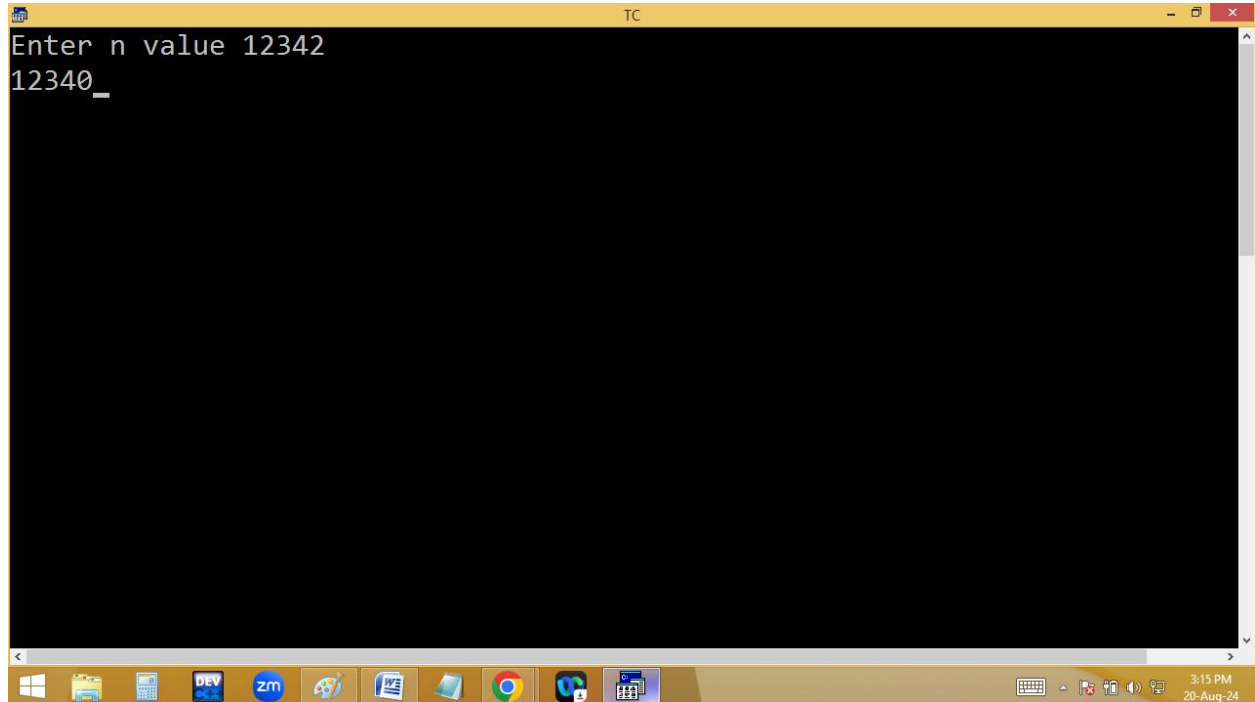
```
#include<stdio.h>
#include<conio.h>
void main()
{
    int n;
    clrscr();
    printf("Enter n value "); scanf("%d",&n);
    n%10>=5 &&printf("%d",(n/10+1)*10)||printf("%d",n/10*10);
    getch();
}
```

The Windows taskbar at the bottom shows the time as 3:14 PM on 20-Aug-24.



The screenshot shows the Turbo C++ (TC) IDE with the program's output. The text "Enter n value 27" is displayed on the first line, and "30" is displayed on the second line. The status bar at the bottom shows the time as 3:14 PM on 20-Aug-24.





```
Enter n value 12342
12340_
```

Increment / Decrement / modify operators[++/--]:

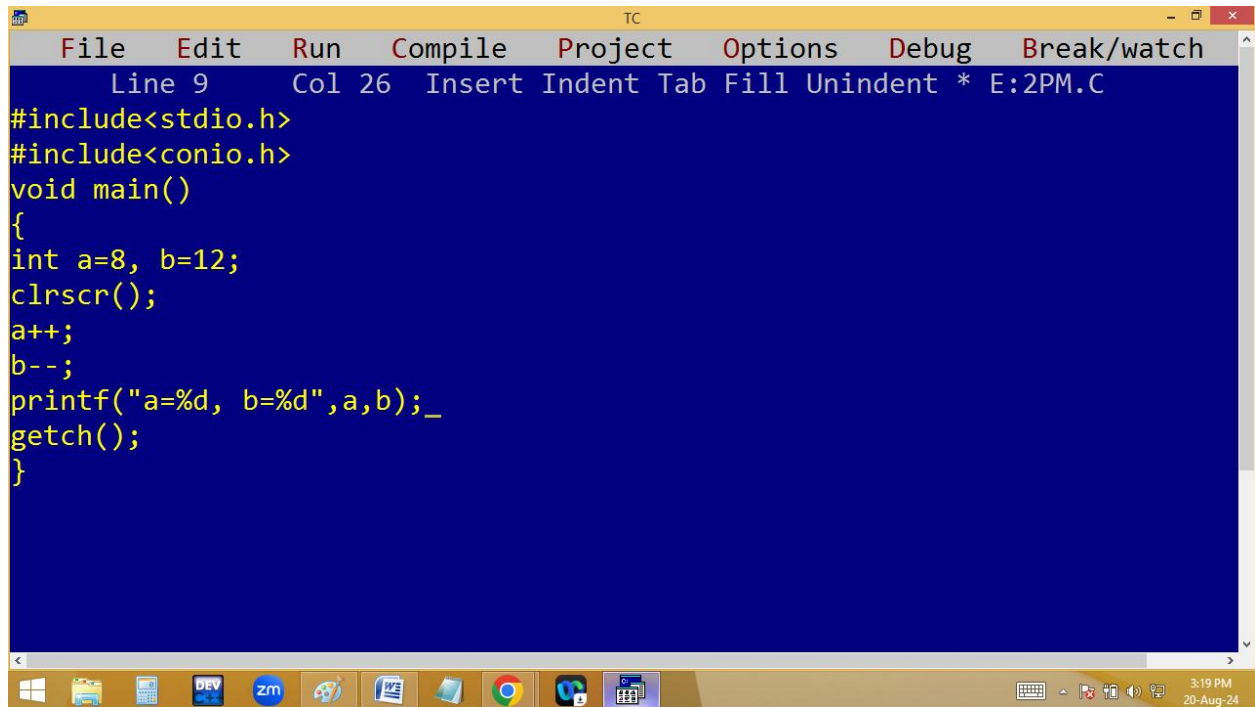
They are used to increment / decrement a variable value by 1.

Eg:

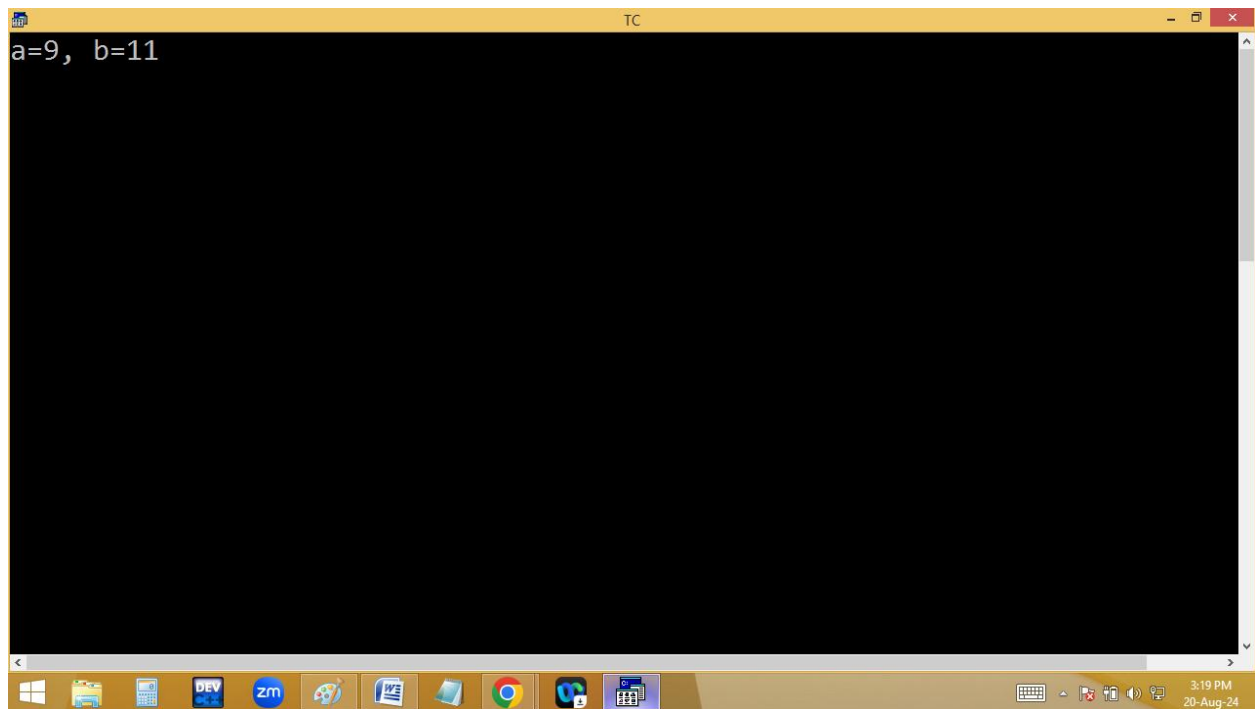
Int a=8, b=12;

a++ i.e. a=a+1 → a=9

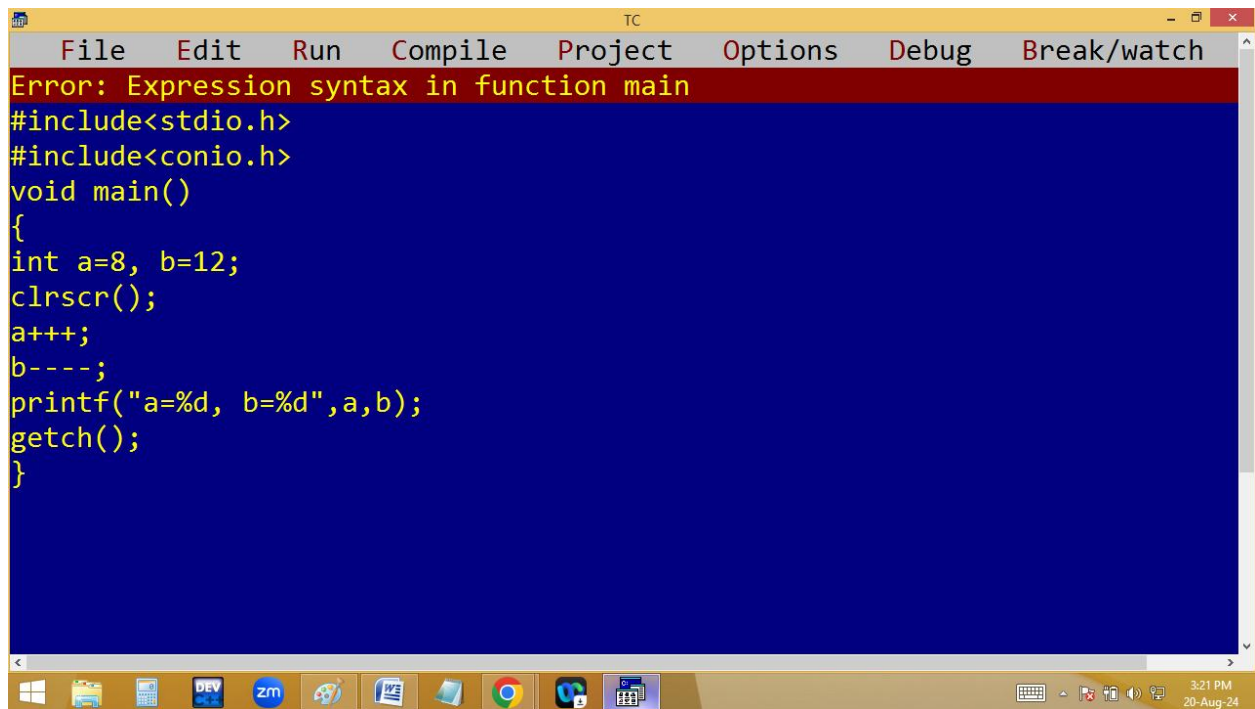
b-- i.e. b=b-1 → b=11



```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 9 Col 26 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=8, b=12;
clrscr();
a++;
b--;
printf("a=%d, b=%d",a,b);_
getch();
}
```



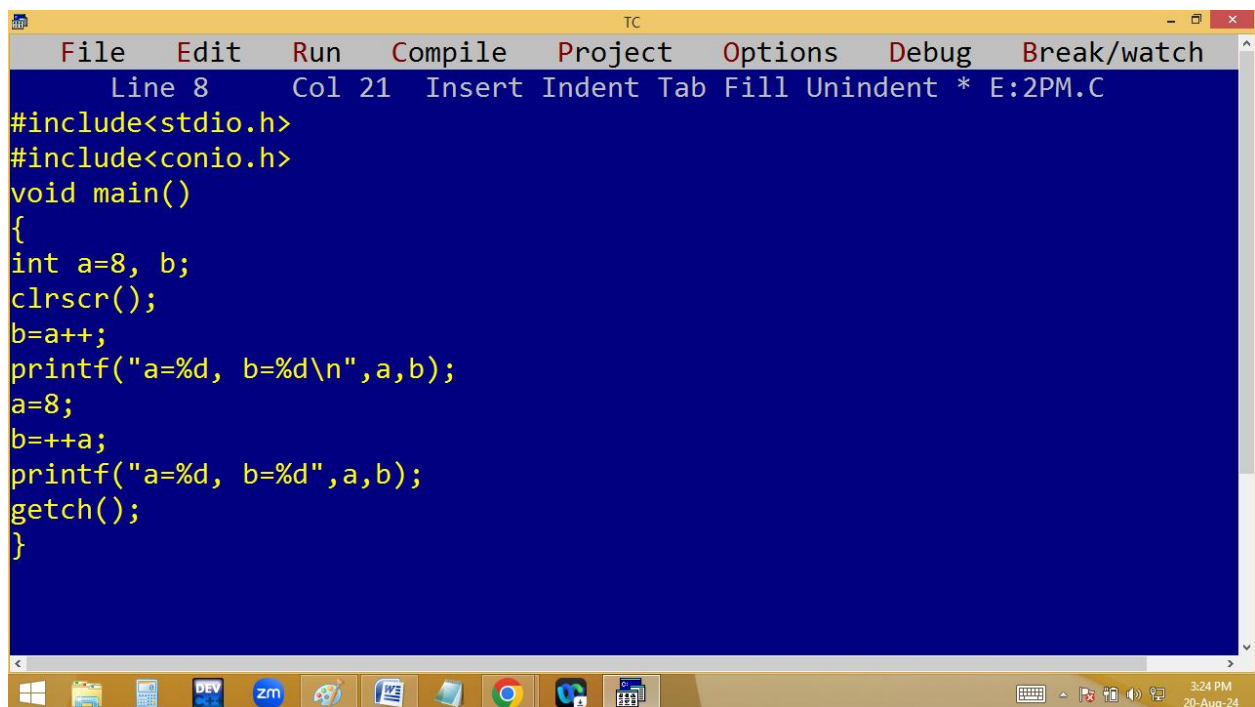
```
TC
a=9, b=11
```



The screenshot shows the Turbo C++ IDE with a menu bar (File, Edit, Run, Compile, Project, Options, Debug, Break/watch) and a dark blue editor area. A red error message bar at the top reads "Error: Expression syntax in function main". The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a=8, b=12;
clrscr();
a++;
b----;
printf("a=%d, b=%d",a,b);
getch();
}
```

The Windows taskbar at the bottom shows the time as 3:21 PM on 20-Aug-24.



The screenshot shows the Turbo C++ IDE with the same menu bar. The editor area now displays the corrected code. At the top of the editor, the text "Line 8 Col 21 Insert Indent Tab Fill Unindent * E:2PM.C" is visible. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a=8, b;
clrscr();
b=a++;
printf("a=%d, b=%d\n",a,b);
a=8;
b=++a;
printf("a=%d, b=%d",a,b);
getch();
}
```

The Windows taskbar at the bottom shows the time as 3:24 PM on 20-Aug-24.

```
TC
a=9, b=8
a=9, b=9_
```

Postfix increment: prefix increment:

a=8
b=a++;
priority: =, a++
1. b=a ==> **b=8**
2. a++ ==> **a=9**

a=8
b=++a;
priority: ++a, =
1. ++a ==> **a=9**
2. b=a ==> **b=9**

Note: Pre is always first and post is always last.

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 10 Col 25 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=8, b;
clrscr();
b=a--; /*posffix decr */
printf("a=%d, b=%d\n",a,b);
a=8;
b=--a; /* prefix decr */_
printf("a=%d, b=%d",a,b);
getch();
}
```

```
TC
a=7, b=8
a=7, b=7_
```

postfix decr:

a=8

b=a--;

priority: =, a--

1. b=a==> **b=8**

2. a-- ==> **a=7**

prefix decr:

a=8

b=--a;

priority: --a, =

1. --a ==> **a=7**

2. b=a ==> **b=7**

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 10 Col 33 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=3, b;
clrscr();
b=a++ + a++ + a++; /*posffix incr */
printf("a=%d, b=%d\n",a,b);
a=3;
b= ++a + ++a + ++a; /* prefix incr */
printf("a=%d, b=%d",a,b);
getch();
}
```

```
TC
a=6, b=9
a=6, b=18
```

a=3

b=a++ + a++ + a++;

priority: +, =, a++

1. b = a + a + a ==> 3 + 3 + 3

2. **b = 9**

3. a++ ==> a=4, a++ ==> a=5, a++==>**a=6**

a=3

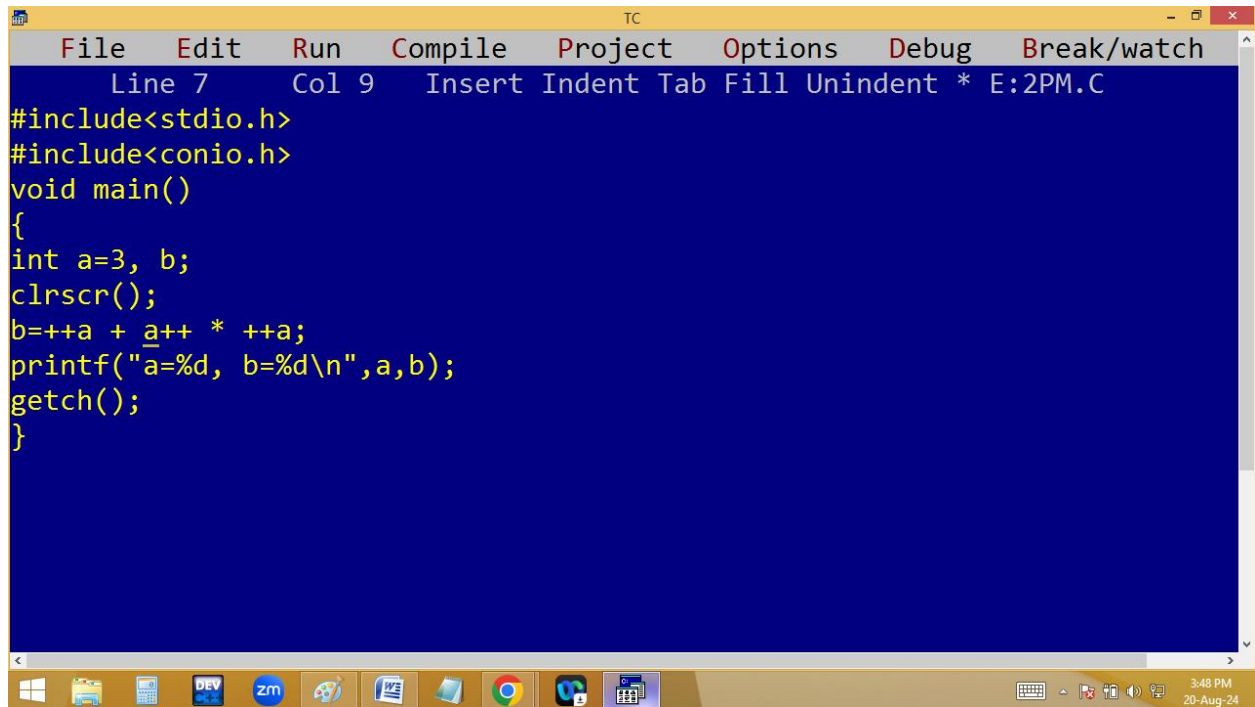
b = ++a + ++a + ++a;

priority: ++a, +, =

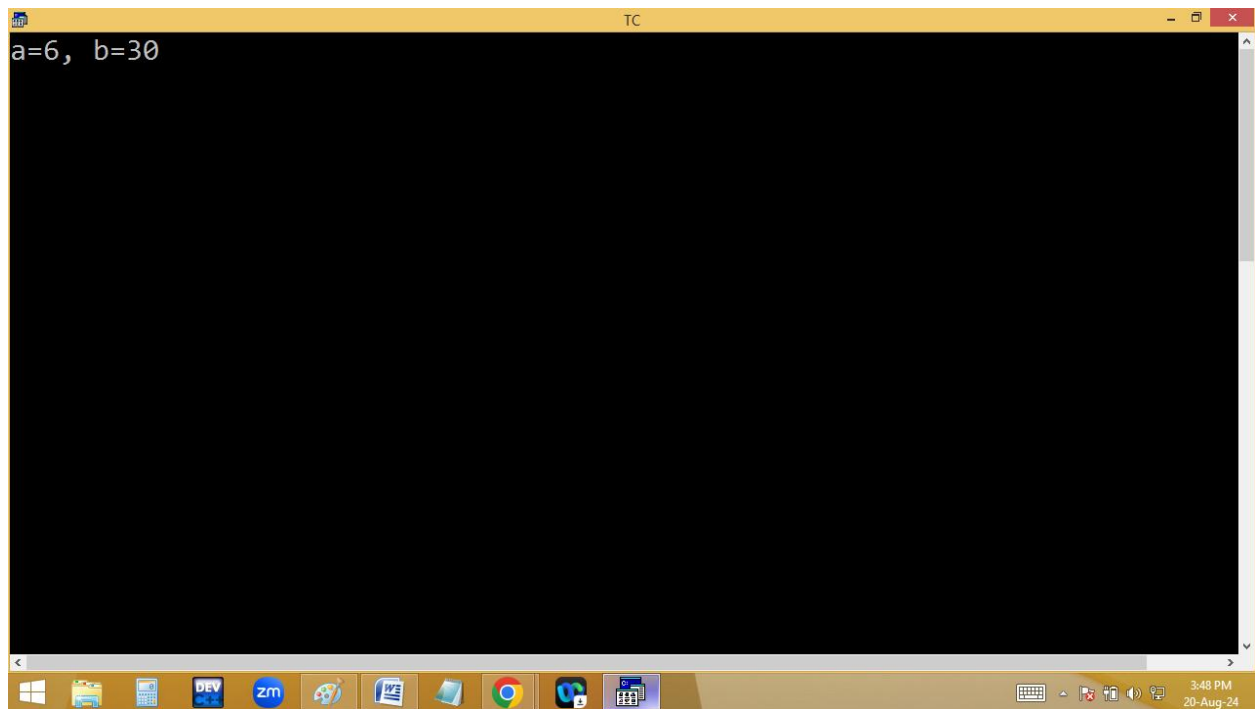
1. ++a==>a=4, ++a==>a=5, ++a==>**a=6**

2. b = a + a + a ==> 6 + 6 + 6

3. **b = 18**



```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 7 Col 9 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=3, b;
clrscr();
b=++a + a++ * ++a;
printf("a=%d, b=%d\n",a,b);
getch();
}
```



```
TC
a=6, b=30
```

a=3

b = ++a + a++ * ++a;

priority: ++a, *, +, =, a++

1. ++a==>a=4, ++a==>a=5

2. b = a + a*a ==> 5 + 5*5

3. b = 5 + 25

4. b=30

5. a++ ==> a=6